V-M2F FRAMEWORK: A MULTIMODAL DEEP LEARNING FRAMEWORK FOR VIETNAMESE CONTENT MODERATION

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Abstract

The rapid growth of online information creates significant challenges for monitoring and moderating harmful or policy-violating Vietnamese content across news portals, social media platforms, and user-generated content sites. Many existing systems still rely on manual filtering or process only a single modality, such as text or images, which limits their effectiveness in practice. In this work, we propose V-M2F, a modular and extensible multimodal framework for Vietnamese content moderation that simultaneously processes both textual and visual information within a unified architecture. Each component — including text processing, image processing, and feature fusion — is designed as an independent, interchangeable module so that different backbone models can be integrated or replaced without affecting the overall system. For multimodal learning, features are extracted separately from text and images using modern language models (ViDeBERTa, XLM-RoBERTa) and vision models (ConvNeXt V2, CoAtNet), and then combined through a lightweight MLP-based fusion mechanism for binary classification. Experiments are conducted on a dataset of 7,000 Vietnamese web content samples to evaluate the effectiveness of the framework. The multimodal approach achieves an accuracy and F1-score of approximately 97.15%, consistently outperforming unimodal baselines. The proposed framework provides a practical and flexible solution that can be adapted to real-world Vietnamese content moderation systems, with clear pathways for incorporating advanced fusion mechanisms and additional modalities in future work.

Index Terms *— Multimodal Learning; Content Moderation; Vision–Language Models; Multimodal Fusion; Vietnamese Language Processing; Toxic Content Detection; Automated Moderation Framework.*

I. INTRODUCTION

The proliferation of user-generated content on Vietnamese web platforms — including news portals, social networks such as Facebook and Zalo, and discussion forums — has dramatically increased the volume of potentially harmful, sensitive, or policy-violating material that must be reviewed and moderated. Manual moderation is no longer scalable at this volume, and automated systems are urgently needed.

Existing automated approaches to content moderation are predominantly unimodal, processing either text or images independently. Text-only systems fail when harmful intent is conveyed through images; image-only systems miss context carried in accompanying captions or article bodies. Real-world web content, however, is inherently multimodal: an image and its surrounding text often only convey a complete meaning together.

This paper proposes V-M2F, a modular multimodal framework that jointly processes text and images from Vietnamese web content. Rather than proposing a novel deep learning model, the primary contribution of this work is a system-level architecture that: (1) enables flexible substitution of backbone models across text, image, and fusion modules; (2) is empirically validated across multiple model configurations; and (3) is designed with practical deployment on Vietnamese web environments in mind.

II. RELATED WORK

A. Text-Based Content Moderation

Text-based content moderation methods mainly focus on analyzing the semantics and syntax of content to detect harmful elements such as offensive language, misinformation, or policy-violating content. Traditional approaches employ natural language processing techniques such as Bag-of-Words and TF–IDF combined with machine learning models such as SVM or Naive Bayes. However, these methods are limited in capturing context and deeper sentence meaning.

The development of deep learning has significantly improved the effectiveness of such systems, especially with the emergence of Transformer-based architectures. Models such as BERT, RoBERTa, and multilingual variants have demonstrated strong

semantic representation capabilities [1], [2]. For Vietnamese, specialized models including PhoBERT [15] and ViDeBERTa [3] have been effectively applied to tasks such as hate speech detection, sentiment classification, and toxic content identification. Nevertheless, these approaches still process text alone and do not leverage complementary information from images.

B. Image-Based Content Moderation

Image-based moderation relies on visual feature extraction to identify explicit, violent, or otherwise harmful visual content. Early approaches relied on handcrafted features such as color histograms or texture descriptors, which proved insufficiently robust for real-world scenarios.

With the advancement of deep learning, convolutional neural networks (CNNs) such as VGG, ResNet, and EfficientNet have been widely used for image classification tasks. More recently, Transformer-based vision models such as Vision Transformer (ViT) and hybrid CNN–Transformer architectures such as ConvNeXt V2 [7] and CoAtNet [5] have shown state-of-the-art performance in visual feature extraction. Similar to text-based methods, however, these systems operate independently and do not incorporate contextual information from accompanying text.

C. Multimodal Deep Learning for Content Moderation

To address the limitations of unimodal approaches, recent studies have focused on multimodal deep learning, which combines information from multiple data sources such as text and images. Common fusion strategies include early (feature-level) fusion, late (decision-level) fusion, and intermediate fusion via attention mechanisms [17], [18].

More advanced approaches such as CLIP [8] leverage contrastive pre-training to align visual and textual representations in a shared embedding space, enabling strong cross-modal understanding. Models such as BLIP [9] and Flamingo [11] further extend this by integrating generative capabilities. However, these large-scale vision-language models are computationally expensive and may be impractical for deployment in resource-constrained Vietnamese content moderation environments. Multimodal systems have been applied to social media content analysis [8]–[12], fake news detection [3], [4], and automated content moderation [13], [14], consistently demonstrating improvements over unimodal baselines.

D. Research Gaps

Despite this progress, several limitations remain. First, Vietnamese-language multimodal datasets for content moderation are scarce, which limits the development and benchmarking of domain-specific systems. Second, most existing multimodal solutions are tightly coupled to a specific model architecture, making it difficult to update or replace components as new models emerge. Third, advanced cross-modal fusion mechanisms — such as cross-attention or contrastive alignment — have not yet been systematically evaluated in the Vietnamese content moderation context.

These gaps motivate the design of a modular, model-agnostic multimodal framework for Vietnamese content moderation — the primary objective of this paper.

III. PROPOSED ARCHITECTURE FRAMEWORK

A. Main Contributions

The main contributions of this paper are as follows:

- We propose V-M2F, a modular multimodal framework for Vietnamese content moderation that decouples text processing, image processing, and fusion components, enabling flexible system design and maintenance.
- We introduce a model-agnostic architecture that supports seamless integration and replacement of different deep learning backbone models without requiring system redesign.
- We construct a multimodal Vietnamese dataset combining textual and visual content from diverse web sources and provide empirical evaluation across multiple model configurations.
- We provide a detailed characterization of the MLP-based fusion module, including its parameter count and inference latency, to substantiate the claim of computational efficiency.

- We identify clear limitations of the current approach — including the use of basic late fusion and a balanced dataset — and outline concrete directions for future work.

B. System Overview

The proposed content moderation system is designed as a multimodal and modular framework to simultaneously process textual and visual data from websites. As illustrated in Figure 1, the overall architecture consists of five components: (1) a data collection module, (2) a text processing module, (3) an image processing module, (4) a multimodal feature fusion module, and (5) a classification and alerting module. Each component is independently designed as a replaceable module, enabling easy upgrading or integration of new models without affecting the overall system.

C. Website Data Collection Module

This module collects and extracts content from web pages, including text (titles, article bodies, and captions) and associated images. Data collection is performed via web scraping techniques or APIs where available. After collection, data are normalized and temporarily stored for subsequent processing. For text, the system removes unnecessary characters, normalizes encoding, and performs sentence segmentation. For images, the system performs resizing and format standardization.

D. Text Processing Module

The text processing module converts textual data into semantically meaningful feature vectors. Depending on system requirements, modern language models such as ViDeBERTa, XLM-RoBERTa, or PhoBERT [15] can be selected and fine-tuned. Processing includes tokenization, embedding, and feature extraction at the sentence or paragraph level using the [CLS] token representation. The output is a fixed-dimensional vector capturing the semantic information of the textual content.

E. Image Processing Module

This module extracts visual features from images using state-of-the-art computer vision models. Architectures such as ConvNeXt V2 [7] and CoAtNet [5] are pre-trained on ImageNet and then fine-tuned on the task dataset. After preprocessing (resizing to 224 × 224 and normalization), the input image is passed through the vision model to obtain a fixed-dimensional feature vector representing visual content relevant to moderation.

F. Multimodal Feature Fusion Module

In this work, we adopt a late fusion strategy in which features extracted from text and image encoders are concatenated and passed through a multilayer perceptron (MLP) for classification. This approach deliberately avoids introducing architectural complexity at the fusion stage, prioritizing system flexibility and the ability to swap backbone models freely.

As detailed in Table V, the MLP fusion module consists of two fully connected hidden layers (1792 → 512 → 128) with ReLU activations and dropout regularization, followed by a softmax output layer with 2 units. The total number of trainable parameters in the fusion module is approximately 960K, which is negligible compared to the backbone encoders (ViDeBERTa: ~180M parameters; ConvNeXt V2-Base: ~89M parameters). The inference latency of the fusion module alone is under 1 ms on a standard GPU, confirming its lightweight character.

We acknowledge that this simple concatenation-based fusion does not exploit cross-modal interactions between specific text tokens and image regions. More advanced fusion mechanisms such as cross-modal attention or contrastive alignment are identified as important directions for future work (Section VI.B).

TABLE V

MLP Fusion Module Architecture

Layer	Input Dim.	Output Dim.	Activation
Concat (Text + Image)	768 + 1024 = 1792	1792	—
FC Layer 1	1792	512	ReLU + Dropout(0.3)
FC Layer 2	512	128	ReLU + Dropout(0.3)
Output Layer	128	2	Softmax

G. Classification and Alerting Module

The fused feature vector is fed into a classifier to predict the content label (normal or violating). The classification result can trigger automated actions such as hiding content, notifying administrators, or logging events into a moderation system. This module is designed to be flexible, allowing adjustment of the number of output classes and training strategies according to practical requirements.

H. Model Extensibility and Replaceability

A key strength of the proposed architecture is its high modularity. Each component can be replaced with different models without changing the overall structure. For example, the text module can be upgraded from ViDeBERTa to a newer model, or the image module can switch from ConvNeXt V2 to a more advanced architecture. The system can also be extended in the future to incorporate additional modalities such as video or audio.

IV. EXPERIMENTAL IMPLEMENTATION

A. Dataset Construction

To evaluate the proposed framework, we construct a dataset of 7,000 Vietnamese web content samples collected from a variety of online sources, including Vietnamese news portals (e.g., VnExpress, Tuoi Tre), online forums (e.g., Webtretho, Otofun), and Facebook public pages. Each sample consists of two components: a text excerpt (title and/or body) and an associated image. Table I summarizes the dataset distribution.

Data are manually annotated by three independent annotators following clear labeling guidelines: content is marked as Violating if it contains toxic language, hate speech, sexual references, or policy-violating material (per Vietnamese cybersecurity regulations [14]); otherwise it is marked as Normal. Inter-annotator agreement was measured using Cohen's kappa ($\kappa = 0.81$), indicating strong agreement. Disagreements were resolved by majority vote.

We note that the dataset is constructed with a balanced class ratio (50:50) to facilitate initial benchmarking of the framework. We acknowledge that this does not reflect real-world class distributions, where violating content is estimated to constitute 1–5% of total volume. Performance metrics on a balanced test set should therefore not be taken as direct indicators of production performance. Evaluation under realistic class imbalance conditions — using metrics such as AUPRC and macro-averaged F1 — is identified as an important direction for future work.

TABLE I

Dataset Distribution

Class	Content Description	Train	Test	Total
Normal	Valid articles, general news, social media posts	2800	700	3500
Violating	Sensitive, toxic, or policy-violating content	2800	700	3500
Total		**5600 (80%)**	**1400 (20%)**	**7000**

B. Data Preprocessing

For text, preprocessing includes case normalization (lowercasing), removal of special characters, word segmentation using RDRSegmenter [3], and encoding according to each language model's tokenizer. Sequences are truncated or padded to a maximum length of 256 tokens.

For images, inputs are resized to 224 × 224 pixels, converted to RGB format, and normalized using ImageNet mean and standard deviation values. Data augmentation is applied during training, including random horizontal flipping, slight rotation (±10°), and brightness/contrast jitter, to increase diversity and reduce overfitting.

C. Training Configuration

Training hyperparameters are summarized in Table II. To mitigate overfitting given the limited dataset size (5,600 training samples), we adopt a partial fine-tuning strategy: the bottom eight layers of each backbone are frozen, and only the top layers

and the MLP fusion head are trained end-to-end. This reduces the number of trainable parameters and regularizes the model. Early stopping with patience of 3 epochs is applied based on validation loss.

TABLE II

Training Hyperparameters

Parameter	Value	Description
Input image size	224 × 224	Standard for ConvNeXt V2 / CoAtNet
Batch size	16	Fits typical GPU memory
Optimizer	AdamW	Suitable for Transformers
Learning rate	2×10^{-5}	Fine-tuning setting; backbone layers partially frozen
Loss function	Binary cross-entropy	Binary classification
Epochs	15	With early stopping (patience = 3)
Dropout	0.3	Applied in MLP to reduce overfitting
Fine-tuning strategy	Partial freeze	Bottom 8 layers frozen; top layers + classification head trained

D. Text and Image Model Configuration

The text processing module is implemented using ViDeBERTa or XLM-RoBERTa, both of which are Transformer-based models pre-trained on large Vietnamese and multilingual corpora. Sentence-level embeddings are extracted from the [CLS] token (768-dimensional for both models). The image processing module uses ConvNeXt V2 Base or CoAtNet-1, initialized with ImageNet-21K pre-trained weights, producing 1024-dimensional feature vectors after global average pooling.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Evaluation Metrics

We evaluate system performance using Accuracy, Precision, Recall, and F1-score. All metrics are reported on the held-out test set (1,400 samples). We note that, as the test set is class-balanced, these metrics reflect performance under equal class priors. Results should be interpreted in this context; future work will evaluate performance under realistic imbalanced conditions.

B. Comparison: Multimodal vs. Unimodal Models

Results in Table III demonstrate that the multimodal model consistently outperforms unimodal baselines. Image-only models achieve F1-scores of 85.32% (ConvNeXt V2) and 86.39% (CoAtNet), reflecting the inherent difficulty of moderation based on visual content alone. Text-only models perform substantially better — 94.12% (XLM-RoBERTa) and 95.30% (ViDeBERTa) — consistent with the strong semantic content present in Vietnamese text. The best multimodal configuration (ViDeBERTa + ConvNeXt V2) reaches 97.15% F1, a 1.85-point absolute improvement over the best text-only baseline. This gain, while modest in percentage terms, demonstrates that visual information provides complementary signal that reduces misclassification in ambiguous cases.

TABLE III

Performance Comparison Between Unimodal and Multimodal Architectures

Architecture	Model	Data	Acc.	Prec.	Rec.	F1
Unimodal (Image)	ConvNeXt V2	Image	85.20%	86.15%	84.50%	85.32%
Unimodal (Image)	CoAtNet	Image	86.50%	87.00%	85.80%	86.39%
Unimodal (Text)	XLM-RoBERTa	Text	94.10%	94.05%	94.20%	94.12%
Unimodal (Text)	ViDeBERTa	Text	95.30%	95.50%	95.10%	95.30%
Multimodal (Proposed)	ViDeBERTa + ConvNeXt V2	Text+Image	97.15%	97.20%	97.10%	97.15%

C. Comparison Across Module Combinations

Table IV reports performance across four text–image model combinations. All configurations exceed 96.5% F1, indicating that the MLP-based fusion mechanism is stable across different backbone choices. The consistent performance across combinations supports the claim of architectural modularity: replacing individual modules does not cause significant performance degradation. The best combination remains ViDeBERTa + ConvNeXt V2 (97.15% F1), while the worst is XLM-RoBERTa + ConvNeXt V2 (96.55% F1), a gap of only 0.60 percentage points.

TABLE IV

Flexibility Comparison of Module Combinations

Text Module	Image Module	Acc.	F1
PhoBERT	Swin Transformer V2	96.85%	96.80%
XLM-RoBERTa	ConvNeXt V2	96.50%	96.55%
XLM-RoBERTa	CoAtNet	96.70%	96.68%
ViDeBERTa	ConvNeXt V2	97.15%	97.15%

D. Discussion of Fusion Mechanism

The MLP-based fusion employed in this work represents a straightforward late-fusion baseline. While effective in the current experimental setting, this approach has a fundamental limitation: by concatenating independent embeddings, it cannot model fine-grained cross-modal interactions — for example, it cannot identify that a specific word in the text corresponds to a specific object region in the image. More expressive fusion mechanisms, such as cross-modal attention (as in VisualBERT or ViLBERT) or contrastive alignment (as in CLIP), are expected to yield further improvements, particularly on ambiguous samples. These are prioritized in our future work.

E. Error Analysis and Limitations

Despite high overall performance, the system produces misclassifications in three recurring error categories: (1) metaphorical or sarcastic text that the model cannot recognize as harmful without broader pragmatic context; (2) visually ambiguous images where violating content is subtle or partially occluded; and (3) cross-modal inconsistency, where text and image convey conflicting signals (e.g., a neutral image accompanying toxic text). These cases highlight the need for more sophisticated fusion mechanisms and richer training data.

Additional limitations include: (a) the small dataset size (7,000 samples), which constrains generalization; (b) the balanced class distribution, which does not reflect realistic deployment conditions; and (c) the lack of comparison against existing Vietnamese multimodal baselines, which remains an important gap to address.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper proposes V-M2F, a modular multimodal content moderation framework for Vietnamese web environments, enabling simultaneous processing of text and images through a unified, extensible architecture. The framework is designed to be model-agnostic, supporting flexible integration and replacement of deep learning backbone models across all processing stages. Experiments on a 7,000-sample Vietnamese multimodal dataset demonstrate that the best configuration achieves 97.15% accuracy and F1-score, outperforming all unimodal baselines. Stable performance across multiple module combinations confirms the robustness of the modular design. The MLP fusion module is shown to be lightweight (≈960K parameters; <1 ms inference latency), making it compatible with real-time deployment constraints.

We explicitly acknowledge that the primary contribution of this work is a system-level framework design, not a novel fusion or backbone architecture. The use of concatenation-based late fusion is a deliberate choice to maximize modularity and reproducibility. Advanced fusion mechanisms are reserved for future investigation.

B. Future Directions

Future work will focus on the following directions: (1) Scaling the dataset to tens of thousands of samples collected from social media and forums, with realistic class-imbalanced distributions, to improve generalization and enable meaningful evaluation under production conditions. (2) Investigating advanced fusion mechanisms, including cross-modal attention, multimodal Transformers, and contrastive representation learning (e.g., mCLIP), to exploit deeper text–image interactions. (3) Extending the architecture to additional modalities such as video and audio. (4) Optimizing the system for real-time deployment with quantified computational cost benchmarks (FLOPs, latency, memory). (5) Integrating the framework into live website management platforms with human-in-the-loop feedback mechanisms.

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